

■ 1.9.4 Splicing *Mathematica* Output into External Files

If you have a C or FORTRAN program, or a T_EX document, you may want to insert formulae or other data by “splicing in” output from *Mathematica*.

<code>Splice["file.mx"]</code>	splice <i>Mathematica</i> output into an external file named <i>file.mx</i> , putting the results in the file <i>file.x</i>
<code>Splice["infile", "outfile"]</code>	splice <i>Mathematica</i> output into <i>infile</i> , sending the output to <i>outfile</i>

Splicing *Mathematica* output into files.

The basic idea is to set up the definitions you need in a particular *Mathematica* session, then run `Splice` to use the definitions you have made to produce the appropriate output to insert into the external files.

```
#include "mdefs.h"

double f(x)
double x;
{
double y;

y = <* Integrate[Sin[x]^5, x] *> ;

return(2*y - 1) ;
}
```

A simple C program containing a *Mathematica* formula.

```
#include "mdefs.h"

double f(x)
double x;
{
double y;

y = -5*Cos(x)/8 + 5*Cos(3*x)/48 - Cos(5*x)/80 ;

return(2*y - 1) ;
}
```

The C program after processing with `Splice`.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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